Hybrid Renewable Energy Generation Solution
Daily Work Report (June 14 – June 20)

June 14 – Login Page Development

The login module was developed for secure access to the Hybrid Renewable Energy Generation Solution. The page includes username/email and password authentication, validation rules, error handling, session management, and user-friendly navigation. The objective is to ensure that only authorized users can access system dashboards and renewable energy monitoring features. Deliverable: Fully functional Login Module with authentication workflow.

June 15 – Registration Page Development

A registration module was created to allow new users, administrators, and energy operators to create accounts. Input validation, password security checks, and database integration planning were included. This module ensures proper user onboarding and account management. Deliverable: Registration Module capable of storing and managing user information.

June 16 – Dashboard Development

The project dashboard was designed and developed to display important information related to renewable energy generation. The dashboard provides summaries of solar and wind power production, system status, energy consumption statistics, and performance indicators. A responsive layout was created to improve usability. Deliverable: Dashboard UI displaying key renewable energy metrics.

June 17 – CRUD Form Development

CRUD (Create, Read, Update, Delete) forms were developed for managing renewable energy data. These forms allow administrators to add new energy sources, update generation records, view stored information, and remove outdated entries. Proper validation and user interaction mechanisms were included. Deliverable: Complete data management forms for renewable energy records.

June 18 – Table & Search Features

Advanced table and search functionalities were implemented to improve data accessibility. Users can view renewable energy records in tabular format, sort information, filter results, and perform keyword searches. These features enhance operational efficiency and reporting. Deliverable: Interactive data listing and search system.

June 19 – Frontend Testing

Comprehensive frontend testing was conducted to verify interface functionality, responsiveness, navigation flow, and form validations. Testing ensured that all modules worked correctly across different screen sizes and user interactions. Deliverable: Frontend Review and Testing Report documenting observations and corrections.

June 20 – Spring Boot Project Setup

The backend environment was initialized using Spring Boot. Project dependencies, configuration files, package structure, and development standards were established. Database connectivity planning and API architecture preparation were also completed, creating the foundation for backend development. Deliverable: Backend Setup with Spring Boot project structure ready for implementation.

Conclusion: The activities completed between June 14 and June 20 established the core frontend and backend foundation of the Hybrid Renewable Energy Generation Solution. The completed work includes authentication, user management, dashboard creation, data handling features, testing, and backend project initialization, preparing the system for further API and database integration.